

DATA SHEET

Automatic 500/1000kN Testing Machines for Steel and Concrete



Automatic Compact-Line combined machine 500/1000 kN, 500 kN for tensile testing on steel rebars mod.70-S12A02

General description

Introduction

This machine has been designed for use, in the field or laboratory to perform tension tests on steel rebars up to 26 mm dia. and compression tests on cylindrical concrete specimens up to dia. 160x320 mm and cubes up to 150 mm, using the appropriate accessories. The system includes the UTM AUTO Power and Control System (PCS) for the automatic execution of compression and tensile tests.

Frame

The frame consists of a very rigid structure with double acting cylinder assembly.

It includes tensile grips and it is fitted with a 150 mm travel high precision displacement transducer allowing test execution under cross-head separation control. See Technical Specifications.

The machine is supplied complete with a set of tensile holders, 4 wedge grips for flats up to 13 mm thickness, 4 wedge grips for rounds up to 26 mm dia. and 2 sets of grip liners 4 and 8 mm thick.

Spherical seat and compression platens for concrete specimens are not included and have to be ordered separately (see Accessories).

UTM AUTO Power and Control System (PCS)

Automatically performs steel tensile and concrete compression tests when connected to the relevant accessories. See Technical specifications.

The UTS Light software (included) is tailored for tensile tests according to EN 6892-1 (method B) and EN 15630-1, being suitable for data acquisition and test results elaboration. See Technical Specifications.

Main features

- Fully automatic test cycle with closed-loop digital feedback
- High precision displacement transducer included, for automatic control of the test and plot of stress/elongation graph, based on the crossheads separation measurement
- No need to use external extensometer to control the test
- Best quality/price ratio
- Heavy duty high-functionality jaws rated for severe prolonged use for testing up to 26 mm dia. rebars
- For tensile tests up to 500 kN on steel and compression tests up to 1000kN on concrete
- Double stage hydraulic pump with rapid approach and precise oil flow control
- Adopts the latest ES technology for reduction of power consumption

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- Data acquisition and test results elaboration to EN ISO 7500-1, EN ISO 6892-1 (method B) and EN 15630-1 by intuitive and very comprehensive software (included)
- Available with stand alone control console or in the integrated Compact-Line version

Standards

- EN ISO 6892
- EN ISO 7500-1
- ASTM A370
- EN ISO 15630-1

Specifications

Frame

Load capacity in tension: 500 kN

Load capacity in compression: 1000 kN

Tensile jaws including: 4 wedge grips for flats up to 13 mm thickness, 4 wedge grips for rounds up to 26 mm dia. and 2 sets of grips' liners 4 and 8 mm thick.

Heavy duty high-functionality jaws rated for severe prolonged use

Maximum distance between grips (tensile mode): approx. 300 mm

Specimen length (tensile mode): approx. 500 mm

Max vertical clearance for compression (with 70-S0012/1): 695mm

Max. ram travel: 150 mm

Distance between columns: 310 mm

Overall dimensions approx (w x d x h): 1000 x 700 x 1750 mm

Weight approx.: 550 kg

UTM AUTO Power and Control System

Hardware and firmware

131.000 points effective resolution

240 x 128 pixels touch screen graphic display

Closed-loop P.I.D.control

2 channels for load sensors

1 channel to measure crosshead separation travel with 150mm high precision displacement transducer (included)

1 channel to measure rebars elongation with extensometers (see Accessories)

digital linearization of the calibration curve (multi-coefficient)

TENSILE TEST:

Automatic test execution under load/stress control and grips separation control (by using the displacement transducer supplied with the machine) with closed loop PID control

Simultaneous display of load, stress and specimen elongation

Possibility to overlap two elongation/stress graphs:

one obtained with the included displacement transducer measuring crosshead separation travel

one obtained with an optional extensometer (coaxial or universal)

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Graphical test data option showing the load/elongation curve

COMPRESSION TEST:

Automatic test execution of compression tests with closed loop PID control

Simultaneous display of load and stress, load/time graph

Hydraulics

Dual stage pump: centrifugal low pressure for fast approach and automatic switching to radial multi-piston high pressure for loading

DC motor 720 W, 50-60 Hz

Maximum working pressure 700 bar

ES Energy Saving technology

Software

UTS Light data acquisition and processing software (included) for tensile test according to EN Standard allowing:

Input of specimen identification, test and name of customer

real time downloading of test data

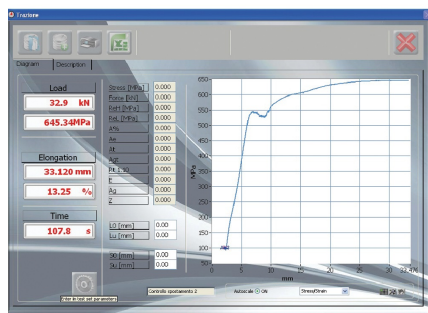
simultaneous display of stress/time and stress/elongation by using coaxial extensometer series 70-C0961/xx or universal extensometer series 70-C0954/x (see Accessories) or displacement transducer reading crosshead separation travel (included)

elaboration of tension test results once test is completed: ReH, ReL or Rp, final elongation, etc. in conformity to EN ISO 6892-1 (method B) and EN 15630-1 (for steel rebars) when extensometer is adopted

unit of measurement: kN, mm, MPa

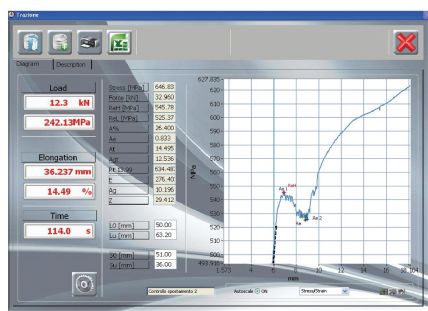
printout of test reports

multi-language software



Test execution, UTS Light software

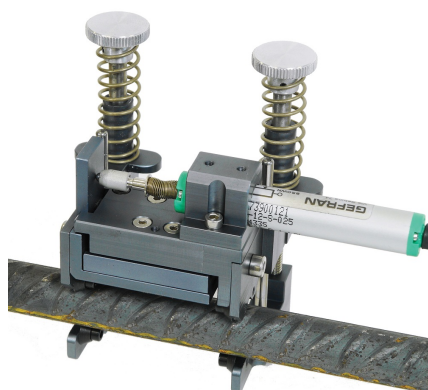
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Test results, UTS Light software



70-C0961/H



70-C0954/C1

Products

70-S12A02



Automatic Compact-Line combined machine 500/1000 kN, 500 kN for tensile testing on steel rebars up to 26mm diameter and 1000 kN for compression testing on concrete. 230 V, 50-60 Hz, 1 ph.

70-S12A04

Automatic Compact-Line combined machine 500/1000 kN, 500 kN for tensile testing on steel rebars up to 26mm diameter and 1000 kN

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for compression testing on concrete. 110 V, 60 Hz, 1 ph.

Accessories and consumables

70-C0954/C1

Electronic universal extensometer to measure the elongation of wires, steel rebars and round steel specimens. To be removed before sample failure. - Measuring base: 50 to 200 mm - Linearity: better than +- 1% - Max. travel: 10 mm

70-S0012/2

Upgrading of the series 70-S11T0x and 70-S12U0x machines with front rigid door

70-C0961/H

Coaxial electronic extensometer for round specimens from 6 to 26 mm dia. Gauge length: 200 mm. Travel: 50 mm.

50-C9080

Distance piece dia 200 x 30 mm

50-C9082

Distance piece dia 200 x 50 mm

50-C9083

Distance piece dia 200 x 68 mm

50-C9086

Distance piece dia 200 x 100 mm.

70-S0012/1

Set of spherical seta and loading platens dia. 220mm for compression test on cylinders up to 160x320mm and cubes up to 150mm. Resulting vertical daylight: 695 mm. Distance pieces not included.

70-S0012/1A

Set of spherical seat with upper loading platen dia.165x30mm and bottom platens dia.250x35 with threaded centering hole. For compression on cylinders up to 160x320. Resulting vertical daylight: 720 mm. Distance pieces not included.